Software Requirements Specification
CSC4351 Capstone 1

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Introduction / Overview

This document outlines the requirements for Elev8, a software application designed to manage and automate elevator operations within a building. The system's primary goal is to efficiently transport passengers via a mobile app between floors by managing requests, optimizing movement, and ensuring secure access.

General description

The Elev8 system is an embedded software component that interfaces directly with elevator hardware and user interfaces.

Functional Requirements

FR 1.0 - User Authentication

FR 1.1: User Login
Description: The system must allow a Passenger to securely log in using their credentials.
Data Inputs: PantherID (Username), Password
Source: Passenger input via mobile app UI fields.
Units of Measure: N/A (Alphanumeric strings)
Range of Valid Inputs: PantherID: Alphanumeric string, 6 to 10 characters. Password: Alphanumeric string, minimum 8 characters.

FR 1.2: ID Verification
Description: The system must verify the entered PantherID against the central user database.
Data Inputs: PantherID, Verification Result (True/False)
Source: Login process (FR 1.1), Central User Database
Units of Measure: N/A
Range of Valid Inputs: Verification Result: {Valid, Invalid}

FR 1.3: Login Error Display
Description: The system must display an error message if the login attempt fails due to invalid credentials.
Data Inputs: Verification Result (Invalid)
Source: ID Verification process (FR 1.2)
Units of Measure: N/A
Range of Valid Inputs: Verification Result: Must be Invalid

FR 2.0 - Core Elevator Calling Operation

FR 2.1: Current Floor Input
Description: The system must allow the Passenger to enter or select their current floor location within the building.

Data Inputs: Current Floor
Source: Passenger input via drop-down list or numeric field on mobile app UI
Units of Measure: Floor Number (Integer)
Range of Valid Inputs: Current Floor: Must be an integer representing a valid floor number, B3 to 20 (or similar predefined range).

FR 2.2: Destination Floor Input
Description: The system must allow the Passenger to enter or select their desired destination floor.
Data Inputs: Destination Floor
Source: Passenger input via drop-down list or numeric field on mobile app UI
Units of Measure: Floor Number (Integer)
Range of Valid Inputs: Destination Floor: Must be an integer representing a valid floor number, B3 to 20, and cannot be equal to the Current Floor (FR 2.1).

FR 2.3: Call Elevator Action
Description: The system must accept the "Call Elevator" request, initiating the service sequence.
Data Inputs: Current Floor, Destination Floor, Action Command (Call)
Source: Passenger interaction: Click/tap on the "Call Elevator" button
Units of Measure: N/A
Range of Valid Inputs: Action Command: Must be Call. Current and Destination Floors must be valid (FR 2.1, FR 2.2).

FR 3.0 - Control System Logic

FR 3.1: Add Current Floor to Queue
Description: Upon a valid call request, the system must add the Passenger's Current Floor to the Elevator Control System's pickup queue.
Data Inputs: Current Floor, Direction Request (Up/Down)
Source: Call Elevator Action (FR 2.3)
Units of Measure: Floor Number
Range of Valid Inputs: Direction Request: {Up, Down}. Direction is determined by comparing Current Floor to Destination Floor.

FR 3.2: Optimize Elevator Operation
Description: The Elevator Control System (ECS) must calculate the most optimal elevator unit and route to fulfill the current floor request.
Data Inputs: Queue of Call Requests (Current Floor, Direction), All Elevator Locations, All Elevator Statuses (e.g., Door Open, Traveling)
Source: ECS backend database and real-time sensor data feed
Units of Measure: Floor Number, Elevator Status (String)
Range of Valid Inputs: Elevator Status: Must be one of {Idle, Moving Up, Moving Down, Door Open, Out of Service}.

FR 3.3: Track Elevator Location
Description: The system must receive and track the real-time location and status of the assigned elevator unit.
Data Inputs: Elevator ID, Current Location (Floor), Status
Source: Elevator Control System hardware/sensors

Units of Measure: Floor Number, Status (Enumerated String)
Range of Valid Inputs: Current Location: Valid Floor Number (B3 to 20). Status: See FR 3.2.

FR 4.0 - Post-Call & Cancellation

FR 4.1: Call Cancellation
Description: The system must allow the Passenger to cancel a pending elevator call request before the elevator arrives.
Data Inputs: Action Command (Cancel), Request ID (The ID of the pending call)
Source: Passenger interaction: Click/tap on the "Cancel Call" button
Units of Measure: N/A
Range of Valid Inputs: Action Command: Must be Cancel. Request ID: Must correspond to an active, unfulfilled call.

FR 4.2: Remove Floor from Queue
Description: The ECS must immediately remove a call request from the pickup queue upon receiving a valid cancellation command.
Data Inputs: Request ID, Action Command (Cancel)
Source: Call Cancellation action (FR 4.1)
Units of Measure: N/A
Range of Valid Inputs: Action Command: Must be Cancel.

FR 4.3: Dispatch Elevator
Description: The ECS must send the finalized destination command to the physical Elevator Control System.
Data Inputs: Elevator ID, Next Target Floor, Direction Command (Up/Down)
Source: Optimize Elevator Operation process (FR 3.2)
Units of Measure: Floor Number
Range of Valid Inputs: Next Target Floor: Must be a valid floor. Direction Command: {Up, Down, Stop}

Interface Requirements

- Call button: Something labeled to request for the elevator.
- Credentials: A way to access the elevator (key card, key forb, QR code, etc.)
- Visual feedback: Clear indicators for the direction of the elevator.
- Touchless operation: Used for first time users that could be accessed with a QR code.
- Accessibility: Designed in a way that the elevator could be used for all users, even with disabilities.

Performance Requirements

- Elevator Response Time: The response time of the elevator getting to the user should be between 0-5 seconds
- App Efficiency: The mobile app should run smoothly while having multiple users
- Memory Usage: The mobile app should not take up more than 100 MB

Design Constraints

- Real time-status update: The app is designed to provide passengers with accurate information on the location, direction, and availability.
- Request handling: The app should handle multiple requests for different floors and directions.
- Call cancelation: Users will have the ability to cancel if they make an accidental floor selection
- Door control: The app has a way to open and close elevator doors for safety purposes.

Non-Functional Requirements

Performance

- Responsiveness: The app must respond to user requests and provide feedback (such as floor indicators)
- Concurrency: The system must handle simultaneous requests from multiple users

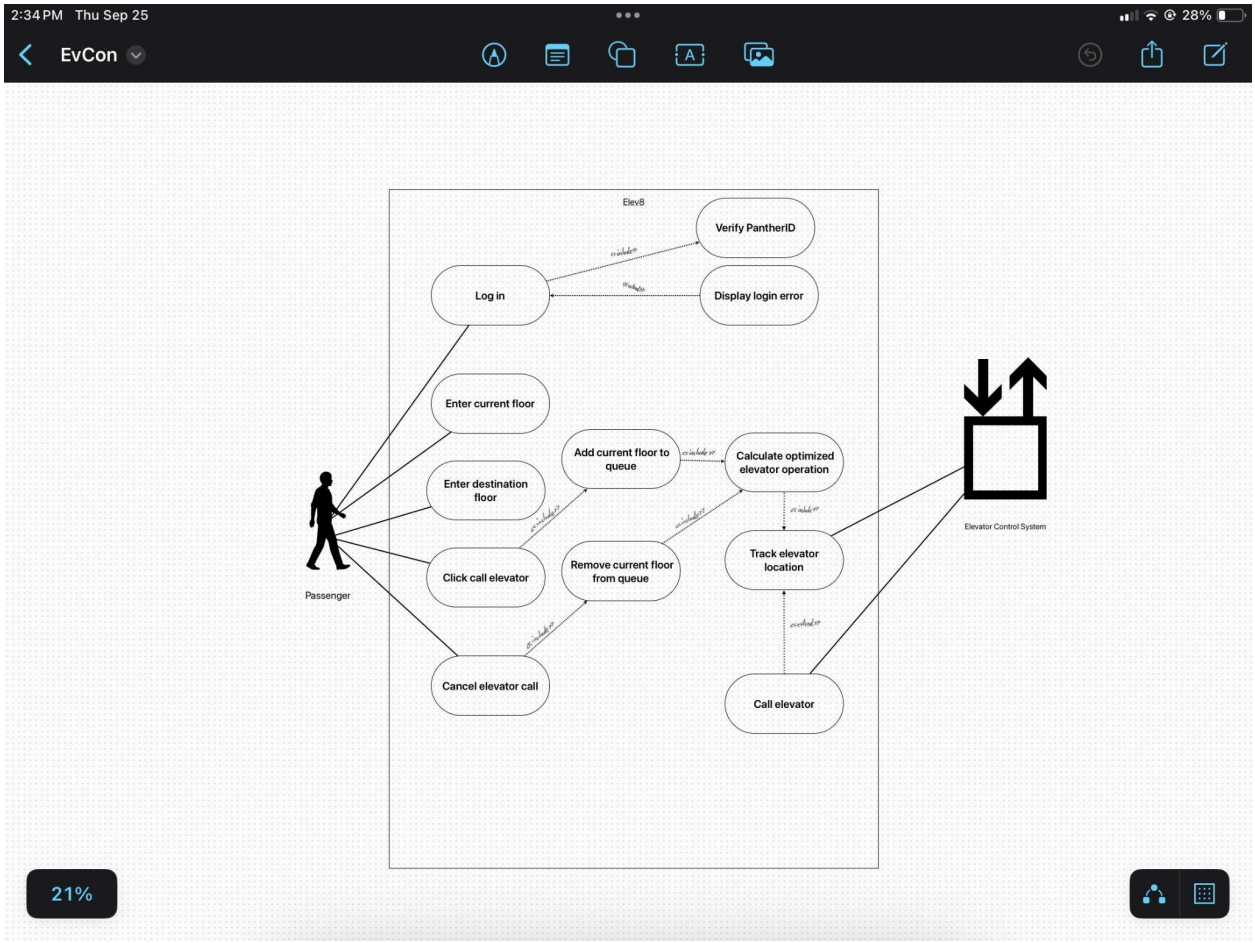
Security

- Authentication: Implement robust authentication mechanisms to ensure only authorized users can access and control the system, preventing malicious use.
- Data Protection: Ensure that any user data collected or transmitted is protected against unauthorized access or tampering.

Reilability

- Availability: The app and its connected elevator system should have high availability, minimizing downtime to ensure constant operation.
- Resilience: Implement recovery mechanisms so that the system can quickly resume normal operation after a disruption.

Appendices


2:34 PM Thu Sep 25
28%
EvCon
Elev8
Verify PantherID
Log in
Display login error
Enter current floor
Add current floor to queue
Calculate optimized elevator operation
Enter destination floor
Click call elevator
Remove current floor from queue
Track elevator location
Cancel elevator call
Call elevator
Passenger
Elevator Control System
21%